Wall time per step: Splat simulation

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

100k (96,342 cells)

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	10	5.8	5.0	5.0	18	2.3	5.1	34	6.1	47	38	9.4	13
qc	0.1	0.1	0.1	0.1	0.6	0.6	0.9	0.5	0.5	1.4	2.6	2.7	1.1
normalise	n/a	n/a	n/a	n/a	2.0	1.9	0.0	0.1	0.1	0.0	0.2	0.2	0.4
hvg	0.3	0.3	0.3	0.2	1.4	1.4	1.4	0.1	0.1	3.8	3.7	3.8	2.9
pca	1.5	1.5	0.6	0.6	2.2	2.2	3.9	1.8	1.8	5.1	11	11	58
harmony	0.5	0.5	0.4	0.3	0.5	0.5	0.5	12	14	14	13	13	14
knn_snn	2.8	2.9	1.7	1.5	27	27	24	6.4	6.4	6.8	7.5	7.6	7.8
cluster	0.9	1.2	0.8	0.8	3.0	2.7	2.7	13	14	13	19	19	21
cluster_fast	1.0	1.0	0.4	0.4	n/a	n/a	n/a	18	18	17	n/a	n/a	n/a
umap	3.8	3.6	0.9	0.7	39	38	38	43	44	43	46	46	46
dge	0.5	0.5	0.5	0.5	47	46	–	3.4	3.5	79	12	11	89
tsne_default	41	42	35	35	162	167	155	n/a	n/a	n/a	125	122	147
tsne_fft	15	15	3.2	2.8	38	38	36	n/a	n/a	n/a	32	32	33
tsne_quick	11	11	3.8	3.5	n/a	n/a	n/a	25	25	23	n/a	n/a	n/a
total (no tSNE)	21	16	10	9.7	141	122	77	115	90	213	152	124	253

500k (482,409 cells)

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	44	28	25	23	OOM	11	21	205	23	225	41	66	
qc	0.3	0.2	0.3	0.3	–	2.8	3.8	2.2	2.2	14	13	4.0	
normalise	n/a	n/a	n/a	n/a	–	3.6	0.0	0.4	0.4	0.9	0.8	1.9	
hvg	0.7	0.8	0.7	0.6	–	3.1	4.5	0.5	0.5	19	19	13	
pca	3.6	3.8	1.3	1.7	–	10	11	14	14	OOM	OOM	271	
harmony	1.7	1.6	1.1	0.6	–	2.1	2.0	100	84	–	–	88	
knn_snn	14	14	7.3	6.6	–	33	29	40	40	–	–	45	
cluster	4.0	4.0	3.3	3.2	–	20	20	116	110	–	–	132	
cluster_fast	14	15	3.1	2.5	n/a	n/a	n/a	196	194	n/a	n/a	n/a	
umap	33	33	3.8	3.5	–	226	226	213	219	–	–	256	
dge	1.8	1.9	1.7	1.7	–	454	–	22	22	–	–	404	
tsne_default	247	245	217	211	–	1,315	1,319	n/a	n/a	–	–	1,404	
tsne_fft	64	61	13	20	–	153	156	n/a	n/a	–	–	157	
tsne_quick	41	41	25	23	n/a	n/a	n/a	155	152	n/a	n/a	n/a	
total (no tSNE)	103	87	45	41	failed	766	317	712	515	failed	failed	1,281	

1M (964,763 cells)

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	97	72	57	44	OOM	OOM	39	OOM	OOM	OOM	OOM	OOM	133
qc	0.5	1.2	0.4	0.4	–	–	7.2	–	–	–	–	–	9.8
normalise	n/a	n/a	n/a	n/a	–	–	0.0	–	–	–	–	–	4.3
hvg	1.4	1.6	1.2	1.1	–	–	8.2	–	–	–	–	–	25
pca	6.6	6.8	2.3	3.1	–	–	18	–	–	–	–	–	468
harmony	3.1	3.0	1.9	1.2	–	–	4.2	–	–	–	–	–	179
knn_snn	29	29	15	13	–	–	40	–	–	–	–	–	100
cluster	8.0	8.4	6.5	6.2	–	–	47	–	–	–	–	–	367
cluster_fast	45	46	8.0	6.2	n/a	n/a	n/a	–	–	n/a	n/a	n/a	
umap	79	78	10	9.8	–	–	555	–	–	–	–	–	555
dge	3.2	3.3	5.4	2.9	–	–	–	–	–	–	–	–	822
tsne_default	641	643	525	526	–	–	TIMEOUT	n/a	n/a	–	–	TIMEOUT	
tsne_fft	195	193	29	45	–	–	–	n/a	n/a	–	–	–	
tsne_quick	136	134	72	73	n/a	n/a	n/a	–	–	n/a	n/a	n/a	
total (no tSNE)	227	204	100	81	failed	failed	719	failed	failed	failed	failed	2,662	

per step: fastest … slowest

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.